

STEADY

Clinical Intelligence Expansion

Execution Overview and Five-Handoff Dependency Map

Clinical Intelligence Expansion | Document 00 | Repository-grounded against tprawlins-lang/Emdr | branch
claude/gifted-keller-501y5d | September 2026

Build directive

Execute five coordinated additions in a strict dependency order so Steady moves from between-visit data collection to functional outcomes, individualized response intelligence, clinician action, longitudinal recovery understanding, and treatment-load decision support without fragmenting its architecture.

Shared product rule

The clinician decides what clinical meaning and action to accept. Steady may measure, organize, compare, and suggest, but it does not silently turn an association, model output, or behavioral signal into clinical fact.

Repository foundation this handoff assumes

Foundation	Current Steady anchor	Rule for this build
Longitudinal history	src/lib/spine.ts + ADR 0010	Durable state must be event-backed and replayable. Do not create a parallel mutable history.
Tenant/person boundary	src/lib/repository.ts + ADR 0011	Every new durable patient record carries tenant_id and person_id; new paths use TenantContext even while legacy paths are migrating.
Clinician patient workspace	src/app/clinician/member/[id]/*	New clinician surfaces extend PersonShell and existing evidence/review conventions.
Evidence-linked summary	src/lib/clinical/summary.ts	A generated claim without resolvable evidence is withheld. Summary is a reading aid, not the record.
Trajectory	src/lib/clinical/trajectory.ts	Separate scales stay separate; provenance remains visible; do not invent a composite clinical meaning from arbitrary axis alignment.
Work queue	src/lib/clinical/work-queue.ts	Clinician home opens on actionable work, stable server ordering, clear reason, one primary action.
Companion memory	src/lib/companion.ts	Patient/Companion data retains source class and memory exposure rules. AI output is not clinician evidence.
AI Gateway	ADR 0012	New model tasks must route through the gateway with schema validation, task version, provenance, evaluation, and PHI policy.

This series also assumes the Clinician Thoughts / Clinical Memory / Session Prep Engineering Specification v2.1 is implemented or being implemented as the common clinician-memory and evidence layer.

Cross-feature invariants

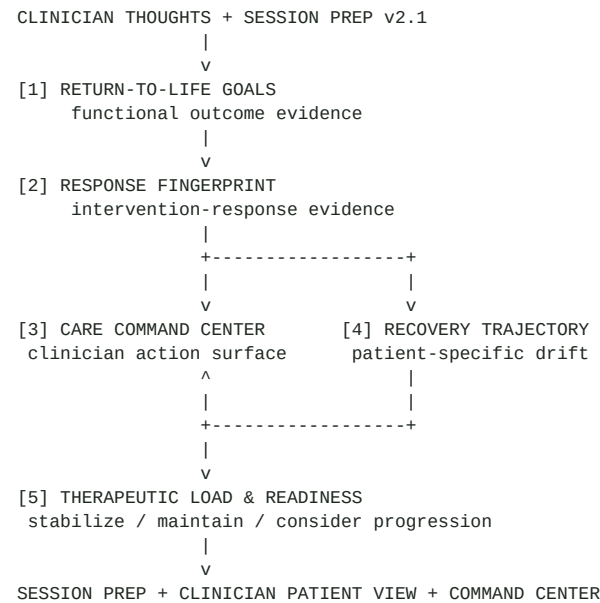
- **Safety authority stays deterministic.** These features may add clinician decision support but cannot clear, weaken, bypass, or replace the safety engine.
- **Stabilization scope stays intact.** No feature autonomously initiates trauma reprocessing, changes a gated treatment decision, diagnoses, or issues a treatment order.
- **Source classes never collapse.** Patient report, clinician observation, system measurement, formal note, Companion interaction, and model inference remain distinguishable.
- **Associations are associations.** A temporal or statistical relationship is not described as causation unless a clinician explicitly records that judgement from appropriate evidence.
- **No future-data leakage.** Historical reconstruction and point-in-time evaluations use only information Steady knew at the relevant cutoff.
- **Missing data stays visible.** Missing observations do not become zero, normal, compliant, or recovered.
- **Corrections append.** An authorized correction supersedes prior derived state without erasing history.
- **Model output is disposable; evidence and provenance are permanent.** Every material intelligence output points to evidence IDs and task/policy versions.

1. The five handoffs and execution order

Order	Handoff	What it adds	Why it comes here
1	Return-to-Life Goals	Individual functional goals and observable progress ladders	Creates the outcome layer. Later systems need to know whether life is actually changing, not just symptoms.
2	Treatment Response Fingerprint	Evidence-linked patterns showing how this person tends to respond to specific interventions in specific contexts	Creates individualized intervention-response data used by later trajectory and load logic.

Order	Handoff	What it adds	Why it comes here
3	Between-Visit Care Command Center	One clinician action surface for non-safety attention signals, follow-ups, goal stalls, response issues, and existing safety work	Creates the operational destination before later engines begin emitting more signals.
4	Personalized Recovery Trajectory	Patient-specific domain trajectories and descriptive deviation detection	Uses goals and response history, then emits explainable attention signals into the Command Center.
5	Therapeutic Load & Readiness	Clinician decision support for stabilize / maintain / consider progression based on recovery burden and capacity	Consumes all prior layers. It must be last because it depends on functional progress, response history, trajectory, and the action surface.

2. Dependency graph



3. Shared intelligence architecture



+--> Patient workspace
+--> Care Command Center

4. Shared runtime contracts

Contract	Owner	Used by
SessionPrepEvidence	Clinician Thoughts v2.1	All five handoffs can expose source-backed content to Session Prep.
Clinical memory items / clinical inferences	Clinician Thoughts v2.1	Goals, response pattern explanation, trajectory explanation, load explanation.
LongitudinalEvent envelope	ADR 0010 / spine	Every durable mutation in all five handoffs.
TenantContext	ADR 0011 / repository	Every patient-scoped read/write/search.
AI Gateway task registry	ADR 0012	Goal drafting, evidence matching, narrative summaries; never safety authority.
Clinical attention signal	Handoff 03	Destination for non-safety actionable signals from handoffs 1,2,4,5.

5. Required execution discipline

- Treat each handoff as an independently mergeable program behind feature flags. Do not start Handoff 2 until Handoff 1 schemas/events are stable; do not start Handoff 4 or 5 before the Command Center signal contract exists.
- Before each phase, inspect the current branch because event-authoritative writes, full TenantContext migration, Postgres cutover, and AI Gateway work are active architecture transitions.
- Every handoff adds its tables to SQLite now and mirrors them to scripts/pg-schema.sql with RLS coverage so future Postgres migration does not create a second redesign.
- Every new signal-producing subsystem must define: evidence inputs, deterministic logic/policy version, limitations, output state, explainability, lifecycle, clinician action, and tests.
- Never add a second clinician home queue. The Command Center extends the existing work queue philosophy and becomes the destination for later actionable intelligence.
- Do not build one global patient score. Steady should preserve domain-specific meaning and explain what changed.

6. Cross-handoff event families

Feature	Core event families
Return-to-Life	return_goal.created / ladder_set / evidence_recorded / progress_confirmed / revised / completed / archived
Response Fingerprint	intervention.exposure_recorded / response_observed / response_pattern_computed / pattern_reviewed
Command Center	attention_signal.opened / updated / acknowledged / action_taken / resolved / reopened
Recovery Trajectory	trajectory.snapshot_computed / deviation_proposed / deviation_resolved
Therapeutic Load	therapeutic_load.snapshot_computed / recommendation_proposed / clinician_reviewed

7. Cross-handoff feature flags

```
RETURN_TO_LIFE_GOALS
RESPONSE_FINGERPRINT
CARE_COMMAND_CENTER_V2
RECOVERY_TRAJECTORY
THERAPEUTIC_LOAD_READINESS
```

```
# AI subfeatures are separately gated
RETURN_GOAL_AI_DRAFT
RESPONSE_PATTERN_AI_SUMMARY
TRAJECTORY_AI_SUMMARY
THERAPEUTIC_LOAD_AI_EXPLANATION
```

8. System-level acceptance test

- A clinician opens one patient and can see: what the person wants back in life, progress toward those goals, interventions and observed response patterns, current longitudinal trajectory, open between-visit attention items, and the latest therapeutic-load recommendation.
- Session Prep can incorporate all five layers without losing provenance or treating AI-generated synthesis as clinician fact.
- The Command Center can show a new goal stall, a trajectory reversal, or repeated recovery burden as separate explainable work items without duplicating safety alerts.
- A cross-tenant ID or semantic retrieval attack cannot reveal any goal, response, trajectory, load, or signal data.
- Replay and point-in-time reconstruction can reproduce all derived clinical intelligence from the evidence and policy versions available at the cutoff.

Appendix A. Senior engineer runbook

- Read this overview, Clinician Thoughts / Session Prep v2.1, ADR 0010, ADR 0011, and ADR 0012 before writing code.
- Execute Handoff 01 through Handoff 05 in order. A later handoff may add providers/consumers to earlier contracts but may not redefine earlier semantics silently.
- For every phase: migration -> domain/events -> projection -> server API -> UI -> security tests -> replay tests -> AI evaluation if applicable -> feature-flag rollout.
- Write or update an ADR for any deviation in source-of-truth, model authority, tenant boundary, or clinical meaning.